Shraboni Sarker

Ph.D. Candidate, Computer Science and Engineering, University of Missouri

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Summary

Machine learning engineer and researcher with five years building multimodal deep learning systems on physiological, behavioral, and clinical data, and four prior years shipping production software to applications with 100M+ downloads. Nine publications (four first-authored) with research funded by NIH, NEH, NSF, PCORI, and the Alzheimer's Association. Equally comfortable designing the model and owning the backend, database, and pipelines it runs on. Graduating December 2026.

Technical Skills

ML & Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, Hugging Face Transformers, CNN, BiLSTM, Transformer, RoFormer, few-shot and low-resource learning, multimodal fusion, time-series modeling, SHAP, attention-based interpretability

Systems & Infrastructure: PyTorch DDP, Ray, Apache Spark, Hadoop, Megatron-LM, vLLM, AWS, Snowflake, Git, Weights & Biases, CI/CD

Languages & Data: Python, SQL, Java, Kotlin, R; relational design, indexing, query optimization, EHR systems (Cerner Millennium, PCORnet Common Data Model), medical ontologies (LOINC, UMLS), CLAMP, wavelet signal processing

Experience

Research Software Engineer, Healthcare AI Jun 2026 – Aug 2026
Department of Physical Therapy, University of Missouri

- Refactored **80% of the MPASS codebase**, consolidating duplicated data-access paths into a single shared service layer — restoring a runnable test suite and shortening the time to add a new assessment type.
- Built and maintained backend services, database architecture, and data pipelines for a digital health platform serving an 83-participant clinical research cohort.

Doctoral Researcher, Scalable Data Science Lab Aug 2021 – Present
University of Missouri (Advisor: Dr. Praveen Rao)

- Designed **CoMo-Former**, a hierarchical CNN–BiLSTM–RoFormer for multi-task postural balance classification: **92.1% AUC** (MCI), 83.2% (frailty), 79.7% (fall risk) across 78 older adults under participant-level GroupKFold, with rotary position encoding contributing a **26.4-point gain** over sinusoidal.
- Built **CRAFT**, a wavelet-based CNN–BiLSTM–Transformer fusing center-of-pressure trajectories with demographics: **97.3% F1** for MCI from a single 20-second standing trial. Published, IEEE/ACM CHASE 2026.
- Fine-tuned M-BERT and BETO on a corpus I constructed, **outperforming ChatGPT-4o few-shot prompting 0.713 vs 0.289 F1** on historical document classification.
- Built a note-classification and ontology-alignment pipeline mapping **132M clinical notes in under two hours** at 73.4% document coverage across a 12-site research network.

Graduate Teaching Assistant Aug 2021 – Present
University of Missouri

- Taught distributed ML and LLM infrastructure (Hadoop, Spark, Ray, PyTorch DDP, Megatron-LM, vLLM) at the 8000 level and database systems (SQL, ER modeling, indexing) at the 3000 level. **Outstanding TA Award, 2026.**

Senior System Engineer Dec 2019 – Jul 2021
Grameenphone (Telenor Group), Bangladesh

- Shipped offers, bundles, and dynamic plan customization for **MyGP (100M+ downloads)** and owned bill-payment and recharge flows for **GPay (1M+ downloads)**, optimizing transaction-heavy features for low-bandwidth networks and a fragmented device fleet.

Earlier: Software Engineer, Silicon Orchard Ltd. (2019); Junior Software Engineer, Media365 Ltd. (2017–2018).

Selected Projects and Funding

MPASS: Mizzou Point-of-care Assessment System NIH/NIA P30AG073105, 2024–2026
Led multimodal deep learning and physiological time-series modeling for a portable screening platform; developed synchronized sensor-acquisition infrastructure supporting an 83-participant cohort. Produced one peer-reviewed publication and one manuscript in preparation.

NSF Innovation Corps (I-Corps) NSF, Spring 2026
*Conducted **130+ customer-discovery interviews** on secure human genomic analysis, applying LLM-assisted qualitative analysis to identify market needs, adoption barriers, and translation opportunities. Received the highest in-person interview recognition.*

SANR: Spanish American Notary Records NEH DHAG HAA-287903, 2022–2024
Designed the annotation protocol with professional paleographers and led corpus release and language-model adaptation for a 220,000-page archive at the Argentine National Archives, producing the collection's first annotated corpus and two publications.

Clinical Document Interoperability PCORI, Greater Plains Collaborative
Built the note-classification and ontology-alignment pipeline for LOINC Document Ontology mapping across a PCORnet network of 12 medical centers in 8 states, supporting standardization of 132M clinical notes.

Aging-in-Place Monitoring Alzheimer's Association, 2024–2027
Contributed LSTM–Transformer research on activities-of-daily-living recognition from non-wearable ambient sensors, investigating personalized temporal modeling for aging-in-place monitoring.

Education

Ph.D., Computer Science and Engineering, University of Missouri Aug 2021 – Dec 2026 (expected)
Dissertation: multimodal deep learning for assessment of mild cognitive impairment, frailty, and fall risk from motor-function data. Advisor: Dr. Praveen Rao. GPA 3.97/4.00

B.Sc., Computer Science (with Distinction), BRAC University, Dhaka Jul 2013 – Dec 2017
Thesis: fall-control system for mobility-impaired users via double inverted pendulum modeling. GPA 3.54/4.00

Selected Publications

Nine publications, four first-authored. Full list on Google Scholar.

- **S. Sarker**, T. M. Guess, P. Rao. "CoMo-Former: A Hierarchical Transformer for Multi-Task Postural Balance Classification." *In preparation, IEEE TNSRE.*
- **S. Sarker**, T. M. Guess, S. Akter, et al. "Deep Learning Classification of Mild Cognitive Impairment Using Static Balance Data." *IEEE/ACM CHASE*, 2026.
- **S. Sarker**, A. T. Hamad, H. Alshammari, V. Grieco, P. Rao. "Seventeenth-Century Spanish American Notary Records for Fine-Tuning Spanish Large Language Models." *ACM/IEEE JCDL*, 2024.
- N. Alrasheed, **S. Sarker**, V. Grieco, P. Rao. "Few-Shot Learning for Word Recognition in Handwritten Seventeenth-Century Spanish American Notary Records." *ACM MM Asia*, 2023.
- **S. Sarker**, P. Rao, M. K. Z. Rana, et al. "Mapping Clinical Notes to LOINC Document Ontology Using EHR Data." *AMIA*, 2022 (poster).
- H. Khan, A. S. M. Mosa, V. Paka, et al., incl. **S. Sarker**. "Mapping Clinical Documents to the LOINC Document Ontology Using EHR Systems Structured Metadata." *AMIA*, 2023.

Honors, Service, and Leadership

- **Graduate/Professional Student Leader of the Year**, University of Missouri (2026); Finalist, TIGERS Achievement Award (2026). **Outstanding Teaching Assistant Award**, EECS (2026).
- **Graduate Excellence Fellowship**, University of Missouri (2021–2022); **1st Runner-Up**, MUIDSI Hackathon, Mizzou / IBM / Show Me Research Week (2024).
- **Peer Reviewer**: IEEE BIBM (2026, Dallas); IEEE Conference on Engineering Informatics (2023); Big Data and Artificial Intelligence (2024, Beijing).
- **Vice President**, College of Engineering Graduate Student Association (2025–2026); **President**, EECS Graduate Student Association (2023–2024); **Invited Panelist**, Engineering Leadership Summit (2026).